Introduction

Gaussian mixture models (GMMs) represent the data as a weighted sum of several multivariate Normal distributions, each component characterising a latent cluster. Unlike k-means, which is essentially a hard-assignment heuristic, GMMs are fully probabilistic generative models: each observation has a posterior probability of belonging to each cluster, the clusters can have different shapes (covariance structures), and model-selection criteria (BIC) handle the choice of cluster count without ad-hoc heuristics. The `mclust` package enumerates a small library of covariance structures and selects the best automatically.

Prerequisites

A working understanding of the multivariate Normal distribution, the EM algorithm for latent-variable models, and the BIC as a model-selection criterion.

Theory

The mixture density is

$$p(x) = \sum_{k=1}^{K} \pi_k \mathcal{N}(x \mid \mu_k, \Sigma_k),$$

with mixing proportions $\pi_k \geq 0$ summing to 1. Maximum-likelihood estimation alternates an E-step (compute posterior responsibilities $\gamma_{ik} = \pi_k \mathcal{N}(x_i \mid \mu_k, \Sigma_k)/p(x_i)$) and an M-step (update π_k, μ_k, Σ_k from the responsibilities). `mclust` parameterises the covariance via 14 structures from spherical-equal-volume to fully-unconstrained-different-shape and orientation, indexed by three letters: volume, shape, orientation, each Equal or Varying. BIC selects the best combination of structure and component count.

Assumptions

Multivariate Normal mixture components, independent observations, and a sample size large enough to estimate component covariances stably (the most flexible models require $\Omega(Kp^2)$ observations for K clusters in p dimensions).

R Implementation

```r
library(mclust)

X <- iris[, 1:4]
fit <- Mclust(X)
summary(fit)

# Soft assignments
head(fit$z)

# Plot
plot(fit, what = "classification")

# BIC across models and K
plot(fit, what = "BIC")
```

Output & Results

`Mclust()` returns the best model code (e.g. "VEV"), the number of components, BIC values across the model space, the fitted parameters, and the matrix of soft assignment probabilities `fit$z`. `plot(fit, what`

= "BIC") shows BIC against component count for each covariance structure, making the selection rationale visible.

Interpretation

A reporting sentence: "mclust selected a VEV (varying volume, equal shape, varying orientation) model with three components (BIC = -560.4, ΔBIC to the second-best model 18.2); the three components closely match the iris species (overall accuracy 97 %), and the soft assignments quantify uncertainty for the seven observations classified differently from their species labels." Always report the chosen covariance structure and BIC value.

Practical Tips

- Use BIC for model selection across K and covariance structure jointly; comparing K at fixed structure or vice versa misses the joint optimum.
- Soft assignments (`fit$z`) are useful for downstream uncertainty propagation; rather than committing to one cluster label, propagate the posterior probabilities into subsequent analyses.
- For high-dimensional data ($p > n$), regularised mixture models (`HDclassif`, `pgmm`) or variational EM are needed; standard `mclust` is unstable when Σ_k has more parameters than observations.
- Outliers can distort the Normal components; consider t-mixture models (`teigen::teigen`) for heavier-tailed components, or trimmed EM (`tclust`) for robustness.
- Log-likelihood can become unbounded for unconstrained covariances when a component covers a single observation; `mclust` adds a small prior to prevent this, but watch for warnings.
- For very large n, mini-batch EM and online variants scale to millions of observations, but the `mclust` defaults handle up to a few thousand comfortably.

R Packages Used

`mclust` for `Mclust()`, BIC selection across the 14-model library, and visualisation; `flexmix` for finite mixtures of regressions and other generalised mixture models; `teigen` for t-mixture models robust to outliers; `tclust` for trimmed mixture models with explicit contamination handling.

For Reviewers

What to look for in a paper using this method.

- **Common misapplications.**
- **Diagnostics that should be reported but often aren't.**
- **Red flags in tables and figures.**
- **What to verify.**
- **What an adequate Methods paragraph must contain.**

See also — labs in this chapter

- PCA, factor analysis, CCA, LDA
- Clustering: k-means, hierarchical, model-based
- UMAP and t-SNE